Note about this PDF: This is the full project documentation; all of this info can also be found in the readme documents within the project.

Start Here! Important Note!

If you haven't yet, unzip the ParayPainter folder somewhere on your computer before continuing.

The easiest way to make new images is to upload the ParayPainter folder to your Google Drive and open the EasyPainter.ipynb file from there. In Google Drive, you simply have to click "New" then "Folder Upload". **Make sure you select ParayPainter only (it will say ParayPrinter in the text entry field)**. Uploading ParayProject would most likely be a waste of time and space (nearly 1GB) on your Drive.

In case you actually do intend to store the images on Drive, **upload the folders separately**, or be prepared to make small changes to the program. I don't want to make any assumptions about your skill level, you may be totally comfortable with editing; the notebook comments should make it easy in that case.

In either case, the next steps are in the readme file in ParayPainter.

Table of Contents

~/ParayProject

- ParayProject/READMEFIRST.txt
- ParayProject/READMEFIRST.md

1 - ParayProject/1950Paintings

No subdirectories. Filenames were generated by GPT-3 models accessed via OpenAI. Contains 1,950 images generated by a customized StyleGAN model, trained from scratch on the artwork of Laura Paray. Although her art is the only imagery the machine has ever seen, and the inspiration clearly shows, it seems to convolute that inspiration into a style of its own. It can't perfectly emulate her style, meaning this tech won't be replacing human artists just yet; still, the progress since 1950 is stunning, isn't it?

2 - ParayProject/ParayPainter

Contains custom model and tool to generate new images. This is the folder to upload to Google Drive.

- BeneaththeBlueSky.png - Sample output used in readme.md
- readme.md - Markdown formatted version of Easy Paray Painter readme.
- readme.txt - Plaintext version of Easy Paray Painter readme.
- **EasyPainter.ipynb - The main attraction.** Contains the executable code necessary to make new images. I often refer to it in these docs as "the tool" or "the program".

2.1 ./db

Short for database.

- nlist.db - Holds name data from GPT-3 for the tool to access. The tool says about 1,950 images can be generated with unique names; to be precise, the number of unique names in the database is 1,963. These names are guaranteed to be unique from each other and from the names in ParayProject/1950Paintings. Altogether, the project contains 3,913 unique names. The more names generated, the less likely they are to be unique. Getting new names from the AI is easy enough, but ensuring the names are truly unique and removing the duplicates is a more labor-intensive process. It's harder the more names we have because there are fewer possibilities. Expanding this database is a possible extension, it would just need to be billed accordingly for the increased data cleaning work.

2.2 ./latentexplore

Save location for images generated with the experimental tool. These images aren't given "artistic" names, just timestamps. The experimental feature is interesting but it needs work and is out-of-scope.

- latent-exploration-notes - Rough notes about the experimental latent vector factorization. The subjectively-identified stylistic commonalities are marked with pluses and minuses, each respectively representing the positive and negative scale used to identify the vector.

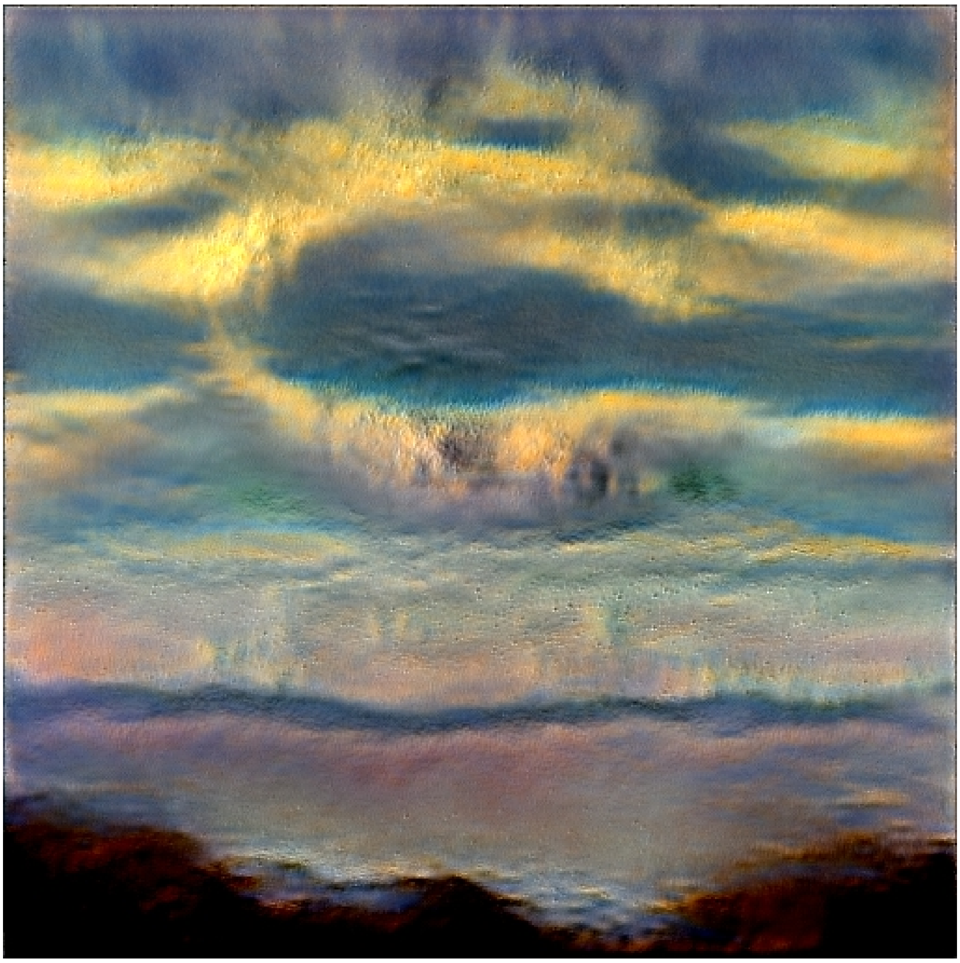
2.3 ./models

- factor.pt - Factorization file used in latent exploration.

- **paray-512px - If the tool is the main attraction, this model is the star.**
  Paray512 was trained on two GPUs for 28 GPU hours; to ensure success, I monitored and adjusted the model approximately hourly. That's fourteen hours real time training, but process took seventeen hours meaning about three hours were spent on adjustments. That also does not take into account the twenty or so models trained for shorter periods, including tests and false starts with ineffective settings. This model was trained from scratch; no prior model was used for transfer learning. In other words, after examining 146 images, and combined with a dose of algorithmic creativity, this AI came up with a style on its own.

Easy Paray Painter

Setup guide for use with free GPU compute from Google Colab



- Step one: Upload this entire folder to your Google Drive. Don't put it in a subfolder or you'll have to make manual changes to the program to run it.

- Step two: From your Google Drive, open the file EasyPainter.ipynb. This should prompt you to open the notebook in Colab.

- Step three: Double-check that you are using a GPU runtime by clicking the dropdown that says "Runtime", clicking "Change runtime type", and selecting GPU in the dropdown if it is not already selected.

- Step four: The notebook should be run cell-by-cell. Notice the [ ] square brackets by the code cells; hover your mouse over and they turn into a play button. Click the play button to run the cell. Check out https://colab.research.google.com/notebooks/basic_features_overview.ipynb for a more detailed overview of Colab.

- Step five: Simply follow the directions in the notebook. The last two cells can be run repeatedly if you want to try different settings. The outputs should save to your Google Drive by default.

Notes about licensing

Some of the code, especially the experimental code relating to closed order factorization, was heavily modified from Kim "rosinality" Seonghyeon's stylegan2-pytorch to work with the Paray512 model. The license for that project is included below.

Arguably enough changes were made that it no longer qualifies as a "substantial portion of the software", but the license is so permissive that it won't hurt to include it. Of course, the "ParayProject" as a whole is unlicensed, all rights are reserved by Paul Paray. Most likely, you're the only one reading this. Replace this paragraph with your own terms as needed, if you plan on any kind of distribution.